

Problem Set 9

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The optimal value of λ for the LASSO model, identified by 6-fold cross validation, was 0.00356. The in-sample and out-sample Root Mean Square Errors (RMSE) were 0.0735 and 0.172, respectively.

For the ridge model, the penalty parameter or λ was equal to 0.0233. In-sample RMSE was equal to 0.0687 and the out-of-sample RMSE was 0.173.

It is not possible to estimate a simple linear regression model when the number of columns is greater than the number of observations or $K > n$. In terms of the bias-variance tradeoff, both models are in the "Goldilocks zone" of the bias-variance curve. Additionally, both models performed similarly on the test or out-of-sample data as they did on the training or in-sample data. However, the fact that the ridge model had a larger penalty parameter yet still had an almost identical RMSE value suggests that it may be the better model. In conclusion, the low RMSE values show that neither models is overfit on the in-sample data.